

JAIDEEP TRIPURANI

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Professional Summary

Full Stack Engineer with **1+ year** of experience building **agentic AI systems**, **RAG pipelines**, and **LLM-powered workflows** backed by a strong **Java Spring Boot** and **Python** backend foundation. At Walmart, integrated **context-aware RAG agents** with **query decomposition**, **multi-step retrieval**, and **FAISS vector search** that automated **25%** of recurring operational workflows. Engineered **Kafka** event-driven microservices processing over **1M events/day** with **99.9% uptime**. Built a production **multi-domain triage agent** using **GPT-4o**, **dynamic task routing**, and **rules-based escalation** — shipped and demoed as part of a national AI hackathon (May 2026) and secured below **100 rank out of 3000**. Excited to build agents that do real work for real users at an AI-first startup.

Work Experience

Full Stack Engineer — Walmart

Oct 2025 – Present

Contract

Vendor: Iconsoft Inc

- Architected and deployed **context-aware RAG pipelines** using **FAISS vector search** and **OpenAI API** implemented **query decomposition**, **query expansion**, and **multi-step retrieval** strategies to improve retrieval precision by **35%** across enterprise inventory knowledge bases.
- Built **agentic AI workflows** with **LangChain** on a **Java Spring Boot** backend agents perform **dynamic task routing**, **result validation**, and **context engineering** across structured and unstructured vendor policy data, automating **25%** of recurring operational workflows.
- Designed **prompt engineering** patterns to improve LLM response accuracy and enforce **schema validation** on agent outputs — reducing hallucination-related failures by **~40%** in production pipelines.
- Engineered **Kafka**-based event-driven pipelines processing over **1 million events/day** implemented consumer group management, partition tuning, and dead-letter queue patterns achieving **40% throughput improvement** with **99.9% uptime** across distributed **Spring Boot** microservices.
- Optimized **PostgreSQL** schemas and **Hibernate ORM** queries — reduced **API response latency by 30%** and improved **query performance by 25%** through targeted indexing and execution plan analysis on high-frequency data access layers.
- Deployed **Spring Boot** microservices on **AWS** (EC2, S3, Lambda, RDS, API Gateway) using **Docker** and **Kubernetes** built **CI/CD pipelines** via **GitHub Actions** and **Jenkins** reducing deployment time by **20%**.
- Built **React.js** frontend features integrated with **Spring Boot REST APIs** across inventory dashboards and supply chain portals maintained **99.9% uptime** serving enterprise users at scale.

Projects

AI Support Triage Agent | *Python, LangChain, FAISS, GPT-4o, Spring Boot, React.js* | May 2026

- Built a production **multi-domain agentic RAG system** for a national AI hackathon (HackerRank Orchestrate) **5 Python** modules with domain-specific **FAISS indexes**, **GPT-4o** generation, **reranking**, **query reformulation**, and **rules-based escalation logic** routing tickets across 3 support domains; **Java Spring Boot** REST backend + **React.js** dashboard deployed on **AWS**.

Global News Market Impact Platform | *Java, Spring Boot, Python, LangChain, React.js, PostgreSQL, AWS*

2024

- Built **Spring Boot** backend with **LangChain RAG pipeline** to retrieve and synthesize market/supply chain news **React.js** frontend, **PostgreSQL** data layer, deployed on **AWS** with **Docker** and **CI/CD** automation.

Technical Skills

Agentic AI & LLMs: LangChain, OpenAI API, GPT-4o, Azure OpenAI, Multi-Agent Systems, Dynamic Task Routing, Tool Calling, Agent Orchestration, Context Engineering, Prompt Engineering, Chain-of-Thought

RAG & Retrieval: FAISS, PyMilvus, RAG Pipelines, Query Decomposition, Query Expansion, Query Reformulation, Reranking (Cross-Encoders), Multi-Step Retrieval, Result Validation, Semantic Search

Backend & Microservices: Java, Spring Boot, Spring MVC, Hibernate/JPA, Python, FastAPI, REST APIs, Kafka, Redis, Event-Driven Architecture, Microservices Design Patterns

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, SQL Indexing, Query Optimization, Schema Design

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS, API Gateway, CloudWatch), Docker, Kubernetes, GitHub Actions, Jenkins, Terraform, CI/CD

Frontend: React.js, TypeScript, JavaScript, HTML/CSS

Testing & Quality: JUnit, Mockito, Unit/Integration Testing, Code Reviews, Schema Validation, Structured Output Enforcement

Certifications: Digital Skills: AI (Accenture), Postman API Fundamentals, Oracle SQL Bootcamp (2025)

Education

University of North Texas

Master of Science in Computer Science

Denton, TX

May 2025

BML Munjal University

Bachelor of Technology in Computer Science

India

May 2023